Chapter 14 worked solutions

DM14-E01

A class a is a unit exactly when $\gcd(a, 8) = 1$. The units are $1, 3, 5, 7$. The remaining nonzero classes $2, 4, 6$ are zero divisors, as witnessed by

$$2 \cdot 4 \equiv 0, \qquad 4 \cdot 2 \equiv 0, \qquad 6 \cdot 4 \equiv 0 \pmod 8.$$

Since $3 \cdot 3 = 9 \equiv 1 \pmod 8$, the inverse of 3 is 3.

DM14-E02

For $p, q \in \mathbb{Z}[x]$, substitution at a preserves addition and multiplication, and reduction modulo 12 also preserves both operations. The constant 1 maps to $[1]$, so this is a unital ring homomorphism. Constant polynomials attain every residue, hence the image is $\mathbb{Z}_{12}$.

Division by the monic polynomial $x - a$ over the integers gives a unique expression

$$p(x) = (x - a)q(x) + p(a).$$

Consequently $[p(a)]_{12} = 0$ exactly when $p(a) = 12k$, and

$$\ker ev_a = \{(x - a)q(x) + 12k : q(x) \in \mathbb{Z}[x],\ k \in \mathbb{Z}\} = (x - a, 12).$$

Here $(x - a, 12)$ is the ideal generated by two elements. A general term $12r(x)$ can be put in the displayed form using $r(x) = r(a) + (x - a)s(x)$. The first isomorphism theorem gives

$$\mathbb{Z}[x]/(x - a, 12) \cong \mathbb{Z}_{12}.$$

With codomain $\mathbb{Z}$, no reduction is taken. The image is $\mathbb{Z}$, and the kernel is exactly $(x - a)$, so

$$\mathbb{Z}[x]/(x - a) \cong \mathbb{Z}.$$

These kernels differ. The constant 12 belongs to the first but not the second. Arbitrarily choosing a representative of a $\mathbb{Z}_{12}$ class does not define an inverse homomorphism to $\mathbb{Z}$, since $[0] = [12]$ would give different integer outputs.

DM14-E03

The Euclidean algorithm gives

$$43 = 2 \cdot 17 + 9, \quad 17 = 1 \cdot 9 + 8, \quad 9 = 1 \cdot 8 + 1.$$

Back substitution gives

$$1 = 9 - 8 = 9 - (17 - 9) = 2 \cdot 9 - 17 = 2(43 - 2 \cdot 17) - 17 = 2 \cdot 43 - 5 \cdot 17.$$

Thus $17^{-1} \equiv -5 \equiv 38 \pmod{43}$, and

$$x \equiv 5 \cdot 38 = 190 \equiv 18 \pmod{43}.$$

DM14-E04

The first congruence gives $x = 1 + 4t$. Modulo 5,

$$1 + 4t \equiv 2, \qquad 4t \equiv 1, \qquad t \equiv 4 \pmod 5.$$

Write $t = 4 + 5s$, so $x = 17 + 20s$. Modulo 7,

$$17 + 20s \equiv 3, \qquad 3 + 6s \equiv 3, \qquad s \equiv 0 \pmod 7.$$

Therefore $x \equiv 17 \pmod{140}$, and the least nonnegative solution is 17.

DM14-E05

The set $I = 2\mathbb{Z}$ is an additive subgroup and absorbs integer multiplication, since $r(2k) = 2(rk) \in I$. It is an ideal of $\mathbb{Z}$.

Likewise, $J = \{[0], [2], [4]\}$ is an additive subgroup of $\mathbb{Z}_6$ absorbing multiplication, hence an ideal. It has 3 elements, so the quotient has $6/3 = 2$ elements. Its cosets are J and $1 + J = \{[1], [3], [5]\}$. Their addition and multiplication agree with residues modulo 2, giving

$$\mathbb{Z}_6/J \cong \mathbb{Z}_2.$$

DM14-E06

The contrapositive of the defining zero-product property of an integral domain is $a, b \neq 0 \implies ab \neq 0$. A counterexample is $\mathbb{Z}_6$, where $[2], [3] \neq [0]$ but $[2][3] = [0]$.

DM14-E07

Since $\gcd(10, 26) = 2$ divides 14, divide by 2 to obtain $5x \equiv 7 \pmod{13}$. As $5 \cdot 8 = 40 \equiv 1$,

$$x \equiv 7 \cdot 8 = 56 \equiv 4 \pmod{13}.$$

The two solutions modulo 26 are $4, 17$. Substitution gives $40 \equiv 14$ and $170 \equiv 14 \pmod{26}$.

For the system, $\gcd(10, 15) = 5$ divides $8 - 3 = 5$, so it is compatible. Writing $x = 3 + 10k$ gives $10k \equiv 5 \pmod{15}$, hence $2k \equiv 1 \pmod 3$, $k \equiv 2 \pmod 3$. Thus

$$x \equiv 23 \pmod{30}.$$

The uniqueness modulus is $\operatorname{lcm}(10, 15) = 30$. Modulo 150, there are five different solution classes, $23, 53, 83, 113, 143$, so uniqueness modulo 150 would be false.

If the second residue is 9, the difference is 6, not divisible by 5, so no solution exists. The first congruence forces residue 3 modulo 5, while the second forces residue 4, a direct contradiction.

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DM14-E08

Additivity forces $\varphi([a]_4) = ae$, where $e = \varphi([1]_4)$, and $4e = 0 \pmod 6$. Multiplicativity forces $e^2 = \varphi(1^2) = e$. Conversely, these two conditions suffice. A change of representative by $4k$ changes the output by $4ke = 0$, proving well-definedness. Addition is preserved, and

$$\varphi(a)\varphi(b) = abe^2 = abe = \varphi(ab).$$

The condition $4e = 0 \pmod 6$ is equivalent to $3 \mid e$, so only $e = 0, 3$ are possible. Both are idempotent, since $0^2 = 0$ and $3^2 = 9 \equiv 3$. Thus all homomorphisms are the zero map and $\varphi([a]) = [3a]$.

The zero map has kernel $\mathbb{Z}_4$ and image $\{0\}$. The nonzero map has kernel $\{0, 2\}$ and image $\{0, 3\}$. Neither sends 1 to 1 of $\mathbb{Z}_6$, so there are no unital homomorphisms. The fact that natural reduction would require $6 \mid 4$ is consistent with this conclusion, but testing only that candidate would not classify all general homomorphisms.

DM14-E09

The set S is nonempty and closed under subtraction and multiplication:

$$\operatorname{diag}(a, 0) - \operatorname{diag}(b, 0) = \operatorname{diag}(a - b, 0), \qquad \operatorname{diag}(a, 0)\operatorname{diag}(b, 0) = \operatorname{diag}(ab, 0).$$

It is a subring. Its own identity is $E = \operatorname{diag}(1, 0)$, which acts as identity on every member of S. The ambient identity $\operatorname{diag}(1, 1)$ is not in S, so it is not a unital subring of the ambient ring. Take $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then $E \in S$, but $AE = A \notin S$. Thus S is not an ideal. Closure only under products within S does not provide ambient absorption.

Since $\det P = 5 - 4 = 1$,

$$P^{-1} = \begin{pmatrix} 5 & -2 \\ -2 & 1 \end{pmatrix}.$$

Its entries are integers, and multiplication with P gives the identity, so P is a unit of $M_2(\mathbb{Z})$. Since $\det Q = 6 - 4 = 2$,

$$Q^{-1} = \frac{1}{2}\begin{pmatrix} 6 & -2 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ -1 & 1/2 \end{pmatrix}.$$

This is an inverse over $\mathbb{Q}$, but not an integer matrix. An integer inverse B would require $\det Q \det B = 1$, giving twice an integer equal to 1, impossible. In general an integer matrix is a unit of its integer matrix ring exactly when its determinant is ± 1. Necessity follows from determinants, and sufficiency follows from the adjugate formula.

DM14-E10

Take $h(a) = a - 5$, with inverse $h^{-1}(y) = y + 5$. Then

$$h(a \oplus b) = a + b - 10 = (a - 5) + (b - 5) = h(a) + h(b),$$
$$h(a \otimes b) = ab - 5a - 5b + 25 = (a - 5)(b - 5) = h(a)h(b).$$

Thus h is an isomorphism to ordinary $\mathbb{Q}$, so the new structure is a field. Its zero is 5, its multiplicative identity is 6, and the additive inverse of a is $10 - a$. Exactly the elements $a \neq 5$ are invertible, with inverse

$$5 + \frac{1}{a - 5}.$$

For $a = 7$, this is $11/2$. Directly,

$$7 \otimes (11/2) = 77/2 - 35 - 55/2 + 30 = 6,$$

the new multiplicative identity. The ordinary numeral 0 is not the new additive zero; it is invertible since $h(0) = -5 \neq 0$.

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DM14-E11

The map $\Theta([x]_{18}) = ([x]_2, [x]_9)$ is well defined and preserves both operations. The residues $9, 10$ map to $(1,0), (0,1)$, respectively, so

$$\Theta^{-1}(a,b) = [9a + 10b]_{18}.$$

Reduction modulo each modulus verifies that this really is the inverse, proving the isomorphism.

An idempotent in the product satisfies $a^2 = a, b^2 = b$ coordinatewise. In $\mathbb{Z}_2$, $a = 0, 1$. In $\mathbb{Z}_9$, $9 \mid b(b-1)$. Consecutive integers are coprime, so the factor 3^2 cannot split between them; it divides one whole factor. Thus $b \equiv 0$ or $1 \pmod 9$. The four pairs $(0,0), (1,0), (0,1), (1,1)$ give exactly

$$x = 0, 9, 10, 1.$$

This proves completeness without enumerating all residues.

Only 1 among them is a unit: cancellation of a unit in $x^2 = x$ forces $x = 1$. The nonzero elements $9, 10$ are zero divisors because $9 \cdot 10 = 90 \equiv 0 \pmod{18}$. By this chapter's convention, 0 is not listed as a nonzero zero divisor.

For a unit in the product, the $\mathbb{Z}_2$ coordinate must be 1, and the $\mathbb{Z}_9$ coordinate can be $1, 2, 4, 5, 7, 8$. There are $1 \cdot 6 = 6$ units. Reconstructing gives $1, 5, 7, 11, 13, 17$, precisely the residues satisfying $\gcd(x, 18) = 1$.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 14.

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